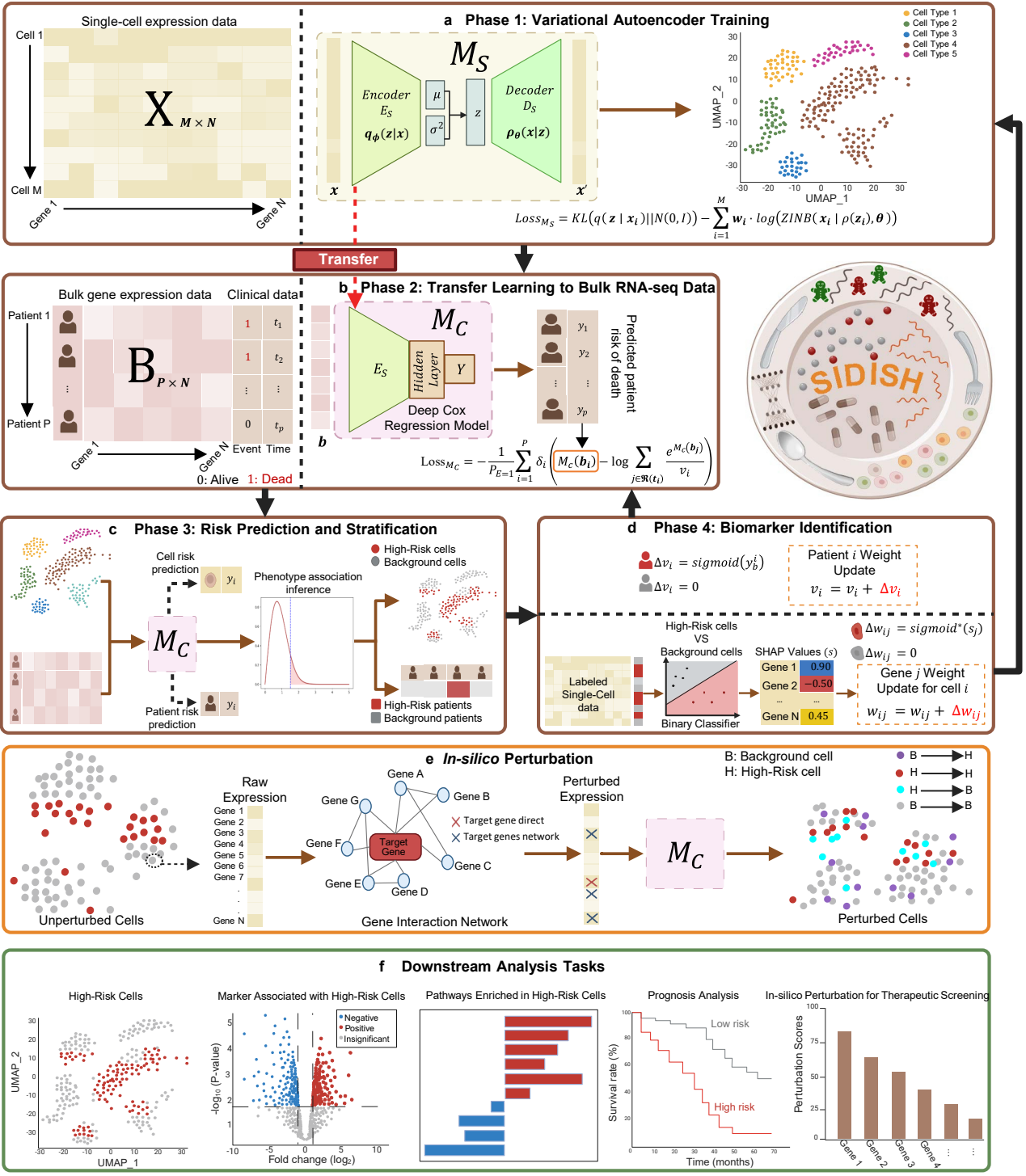
Single-cell expression data
Cell 1
Cell M
Gene 1
Gene N
a Phase 1: Variational Autoencoder Training
Encoder
Decoder
Cell Type 1
Cell Type 2
Cell Type 3
Cell Type 4
Cell Type 5
UMAP_1
UMAP_2
Transfer
Bulk gene expression data
Clinical data
Patient 1
Patient P
Event Time
0: Alive
1: Dead
b Phase 2: Transfer Learning to Bulk RNA-seq Data
Hidden Layer
Deep Cox Regression Model
Predicted patient risk of death
SIDISH
c Phase 3: Risk Prediction and Stratification
Cell risk prediction
Phenotype association inference
High-Risk cells
Background cells
Patient risk prediction
High-Risk patients
Background patients
d Phase 4: Biomarker Identification
Patient i Weight Update
Labeled Single-Cell data
High-Risk cells VS Background cells
Binary Classifier
SHAP Values (s)
Gene 1 0.90
Gene 2 −0.50
Gene N 0.45
Gene j Weight Update for cell i
e In-silico Perturbation
B: Background cell
H: High-Risk cell
Raw Expression
Gene Interaction Network
Target gene direct
Target genes network
Target Gene
Perturbed Expression
Unperturbed Cells
Perturbed Cells
f Downstream Analysis Tasks
High-Risk Cells
Marker Associated with High-Risk Cells
Negative
Positive
Insignificant
Fold change (log2)
Pathways Enriched in High-Risk Cells
Prognosis Analysis
Low risk
High risk
Survival rate (%)
Time (months)
In-silico Perturbation for Therapeutic Screening
Perturbation Scores